

Title: Use-Case Flow Specification**Subtitle:** Problem Statement #44 — Database Query Performance Profiler**NAME :** Sujal Sachin Yadavi**SRN :** PES1UG24CS475**Info table:**

Field	Value
Use Case	Generate Weekly Optimization Digest
Use Case ID	UC-04
Actors	Backend Lead (primary), Database Administrator, Notification Service (secondary)
Includes	«include» Identify Missing Index Candidates (UC-03)
Preconditions	1) At least one week of slow query logs has been ingested (UC-01). 2) The scheduled digest job is configured and enabled. 3) Backend Lead and Database Administrator have valid notification subscriptions (email/Slack).
Postconditions	Success: A digest document is generated, persisted, and delivered to all subscribed stakeholders, and its delivery status is logged. Failure: No digest is delivered; the system logs the failure reason and retries or notifies the Backend Lead.

Main Success Scenario

1. Backend Lead's configured weekly schedule triggers the digest job, or the Backend Lead manually requests a digest from the dashboard.
2. System retrieves all slow query log entries ingested during the past 7 days for the configured database instance(s).
3. System invokes Identify Missing Index Candidates («include», UC-03) to parse execution plans and generate index recommendations for the period's slow queries.
4. System ranks queries by cumulative impact (frequency × average duration) and selects the top slow queries and their associated recommendations.
5. System compiles the digest, including summary statistics, the top slow queries, recommended indexes, and estimated performance impact for each.
6. System persists the digest record and marks it with a generation timestamp.
7. System sends the digest to the Notification Service, addressed to the Backend Lead and Database Administrator.
8. Notification Service delivers the digest via email/Slack and returns a delivery confirmation.

9. System logs the successful delivery status against the digest record.

10. Use case ends successfully.

Alternate Flow

7a. Notification Delivery Fails

- 7a. Notification Service returns a delivery failure (e.g., invalid address, service outage).
- 7b. System logs the failure reason and retries delivery up to 2 additional times, with a 15-minute interval between attempts.
- 7c. If all retries fail, the system marks the digest as "Delivery Failed," keeps the digest available on the dashboard, and sends an in-app alert to the Backend Lead.

6a. No Slow Queries Logged in the Period

- 6a. System finds zero qualifying slow query log entries for the past 7 days.
- 6b. System generates a digest indicating "No slow queries detected this week" instead of a ranked list.
- 6c. Use case resumes at step 6 (persist and deliver the digest) and ends successfully.